NISHCHAL BIDADIKAR NATESH

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EDUCATION

Master of Science in Engineering in Automotive and Mobility Systems Engineering — **May 2025**
University of Michigan, Dearborn, MI, USA — *GPA:3.60/4.00*

Bachelor of Technology in Mechanical Engineering — **August 2022**
RNS Institute of Engineering (Visvesvaraya Technological University), Bengaluru, IN — *GPA:3.65/4.00*

SKILLS

Technical: CAN, Triaging, Jenkins, ISO 26262, SOTIF, ASPICE, ASIL, JIRA, MATLAB, Simulink, Power BI, Python, GitHub, TestRail.

EXPERIENCE

Software Verification & Validation Engineer| KPIT Technologies Inc.| *Michigan, United States* — **June 2025 – Present**

- Execute and monitor automated OTA regression test suites for vehicle systems, ensuring reliable validation of software updates across multiple ECU configurations.
- Triage failed and pass with issues runs by analyzing logs, timestamps, and CAN data to identify root causes across software, bench, network, and architecture layers.
- Document defects and test outcomes in TestRail and other test management tools, providing clear, actionable reports for development and stakeholder teams.
- Collaborate with developers and cross-functional teams to clarify issues, align on fixes, and improve overall OTA test coverage and quality.
- Contribute to continuous improvement of verification workflows by refining test scenarios, reporting practices, and automation usage (e.g., Jenkins).

Mobility Systems Integration Coordinator| Invest UP | *Michigan, United States* — **October 2024 – June 2025**

EV Charging Network Expansion & Infrastructure Optimization

- Developed and maintained interactive dashboards using Power BI to illustrate EV charger deployment progress, track key performance metrics, and identify high-priority locations for network expansion.
- Proposed and validated an alternative energy solution for EV chargers, reducing reliance on the grid and lowering operational costs by 20%. Presented findings to the Urban Planning Committee, securing approval and initiating efforts to obtain a federal grant for implementation.

Electrification of Recreational Trails – Feasibility Study & System Design

- Led a feasibility study to evaluate the electrification of recreational trails, developing a comprehensive evaluation criterion and employing a Pugh matrix to compare design options against key *criteria*. Applied a systems approach to ensure alignment with sustainability and user experience goals.
- Prepared findings for presentation at the U.P. Outdoor Recreation Summit, integrating the study into the Outdoor Recreation Innovation Growth Strategy by collaborating with internal and external stakeholders to ensure clear and effective communication.

Automotive Intern| KPIT Technologies Inc.| *Michigan, United States* — **June 2024 – August 2024**

- Conducted comprehensive research on On-Board Diagnostics (OBD) systems, including standards, regulations, and emission norms, to support compliance with EPA, CARB, NHTSA, and UNEP.
- Analyzed and compared powertrain application regulations across various geographies, providing actionable insights for regulatory compliance, certification processes, and regulatory applications (CAFE, GHG)

ACADEMIC PROJECTS AND COURSEWORK

Research TITLE - Reliability and Verification in ADAS and AV Systems |University of Michigan Dearborn — **January 2024 – April 2024**

- Acquired knowledge of essential approaches to ensure reliability and safety in Autonomous Driving Technology by understanding the technical requirements and operational design domain (ODD) of the system.
- Explored key tools such as Failure Mode and Effects Analysis (FMEA), Hazard Analysis and Risk Assessment (HARA) under ISO 26262, and Safety of the Intended Functionality (SOTIF/ISO 21448) in the context of automotive safety.

Automobile - An Integrated System Term Projects| University of Michigan Dearborn — **August 2023 –December2023**

- P1: Executed a benchmarking study to derive system and vehicle-level requirements for a hypothetical My-Hyundai Santa Cruiz 2029, optimizing efficiency, cost, weight, and manufacturability. Proposed a hybrid engine design to enhance energy efficiency, aligning with sustainable automotive solutions.
- P2: Applied Quality Function Deployment (QFD) to translate customer needs into functional specifications for a driver interface system, cascading sub-attribute requirements and conducting interface analysis to ensure system integration.
- P3: Formulated a detailed program timing plan using the systems engineering "V" model, ensuring timely deliverables, and gained hands-on experience in breaking down complex technical requirements into actionable specifications to validate system performance, ensuring alignment with vehicle-level requirements.

KEY ACHIEVEMENTS

- ➢ **Awarded the Michigan Mobility Fellowship - MMF**
- ➢ **Secured approval and initiated efforts to obtain a federal grant for EV charging station expansion in Michigan**